

Junsu Lee

Computer Science Senior, CWRU | AI Systems, Robot Deployment & Data Engineering

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Education

Case Western Reserve University

B.A. in Computer Science - GPA: 3.4

Cleveland, OH

Aug 2020 - Dec 2026 (expected; incl. 18-month military service)

Skills

Programming & Data: Python, SQL, Bash, Pandas, NumPy, Tableau

Physical AI & Robotics: NVIDIA GR00T N1.5, LeRobot, Boston Dynamics Spot (Orbit), SO-101, Teleoperation

Model Training & Operations: Hugging Face, Weights & Biases, NVIDIA Brev, Modal

Backend & Infrastructure: FastAPI, MSSQL, PostgreSQL, Snowflake, Docker, Apache Airflow, dbt

Web: React, Next.js, TypeScript

Hardware & Prototyping: SolidWorks (CSWA Certified), Autodesk Fusion 360, 3D Printing, Hardware Integration

Languages: English (fluent), Korean (native)

Experience

Advanced Manufacturing Engineer - Co-op

Bendix Commercial Vehicle Systems

Avon, OH

Jan 2026 - Present

- Implemented a Boston Dynamics Spot inspection program, translating maintenance requirements into 450+ SAP-aligned inspection points.
- Built a Python/MSSQL integration layer that normalizes 11 Orbit API resource types into a 16-table schema and maps robot inspection data to the SAP asset hierarchy.
- Designed light-and-sound behaviors that signal robot state and direction, turning shop-floor feedback into a production feature.
- Fabricated 120+ 3D-printed AprilTag plates for Spot localization, reducing 3d modeling time from ~2 hours to 2 minutes and unit cost from approximately \$16 to \$2 per tag.

Weather AI Research Intern

xLab @ CWRU — Industry Partner: Progressive

Cleveland, OH

Jun 2025 - Aug 2025

- Developed an HPC weather-model pipeline with Python and NVIDIA Earth2Studio, generating 7-day global forecasts in under 3 minutes.
- Evaluated FourCastNet, StormCast, CorrDiff, and cBottle for extreme-weather forecasting, presenting tradeoffs to stakeholders.

Frontend Developer - AI Call Center Training System

xLab @ CWRU — Industry Partner: Eaton

Cleveland, OH

Jan 2025 - May 2025

- Built frontend components with Next.js, TypeScript, and Material UI for an AI call-center training platform (FastAPI backend).

Squad Leader / IT Technician (Signaler)

Republic of Korea Army

South Korea

Jan 2023 - Jul 2024

- Designed a CCTV bench-testing process cutting install time 70%; trained a five-person team maintaining 400+ networked devices.

Projects

Language-Guided Bimanual Manipulation with NVIDIA GR00T N1.5

CWRU Senior Capstone | Jul 2025 - Dec 2025

- Engineered a 12-DoF bimanual SO-101 embodiment in LeRobot with leader-follower teleoperation and a three-camera capture stage.
- Collected 360 synchronized teleoperation episodes across 12 configurations after diagnosing timestamp misalignment and discarding approximately 200 corrupted episodes.
- Fine-tuned GR00T N1.5-3B on H200/L40S GPUs, raising task success from 0% untuned baseline to ~25%
- Cut training time from a projected 15 hours to 2.5 by profiling video decoding and tuning batch size and data-loader parallelism.

Environmental & HR Data Pipelines (CSDS 397: Data Pipelines & Visualization)

Jan 2025 - May 2025

- Architected a Python/Airflow pipeline automating environmental data ingestion, producing interactive Tableau air-quality dashboards.
- Engineered dbt ELT transformations to cleanse and validate HR datasets, deploying 6 tested models with automated refresh.

Leadership & Awards

Vice President | CWRU Association of Computing Machinery Student Chapter

Aug 2025 - Present

- Hosted a hands-on OpenClaw agentic-AI meetup, guiding ~8 attendees through setup and deployment on Oracle Cloud instances.
- Organized a chapter trip to NVIDIA GTC Washington, D.C. (Oct 2025), bringing 12 students to the conference.

Excellence Prize (3rd), Asan Doers Startup Bootcamp | Asan Nanum Foundation

Aug 2024

- Developed a business model and prototype as product developer, validated through user interviews.

Certifications

- NVIDIA-Certified Associate: Generative AI LLMs** | NVIDIA — Oct 2025
- Deep Learning Specialization** | DeepLearning.AI — Mar 2024